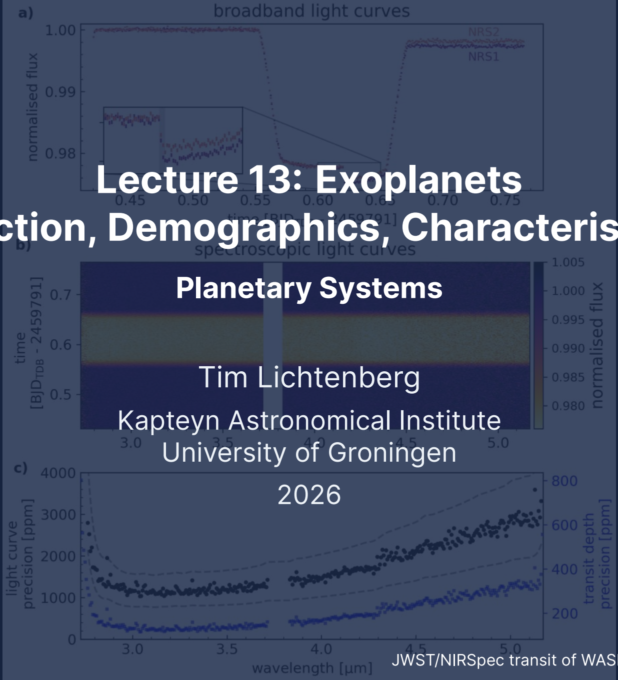
Lecture 13: Exoplanets
Detection, Demographics, Characterisation

Planetary Systems

Tim Lichtenberg

Kapteyn Astronomical Institute
University of Groningen

2026



JWST/NIRSpec transit of WASP-39 b. Credit: Alderson et al. (2023)

Learning Objectives

By the end of this lecture, you will be able to:

- Describe the main exoplanet detection methods and their biases
- Combine transit and radial-velocity geometry into a bulk density, and read the radius valley
- Evaluate JWST-era atmospheric results and biosignature claims

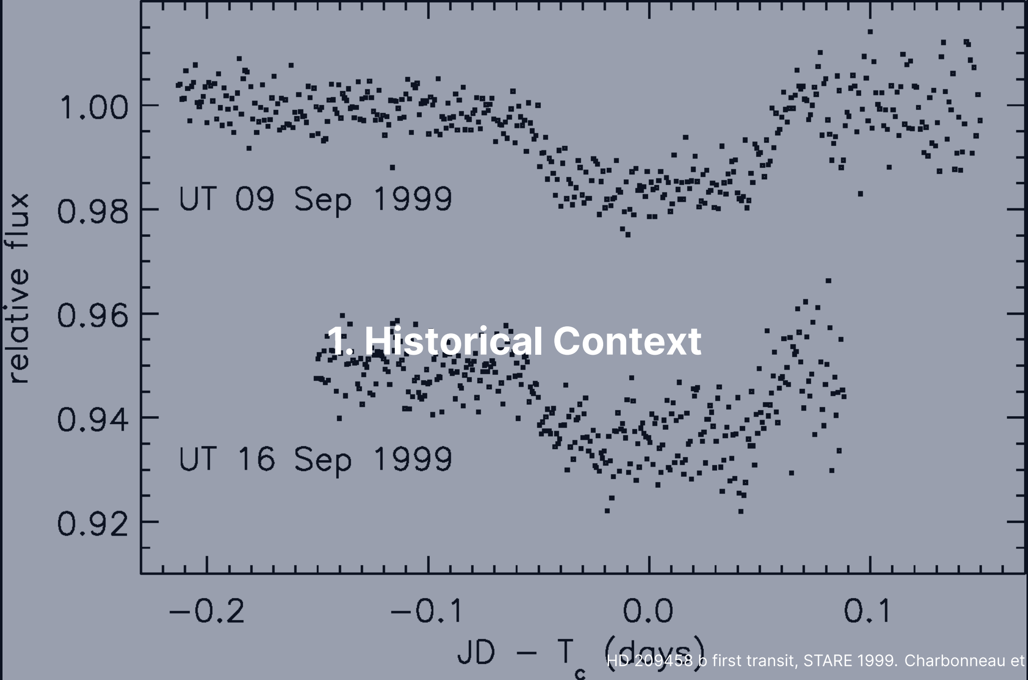
From zero confirmed exoplanets in 1991 to more than 6000 in 2026.

Recap: Where We Are in the Course

- Lectures 9–11 surveyed the geology, atmospheres, and interiors of the solar planets
- Lecture 12 decoded small bodies and meteorites as planet formation fossils
- This lecture turns outward to thousands of exoplanets to test if these patterns are universal

Today: How do we detect, weigh, and characterise worlds beyond our solar system?

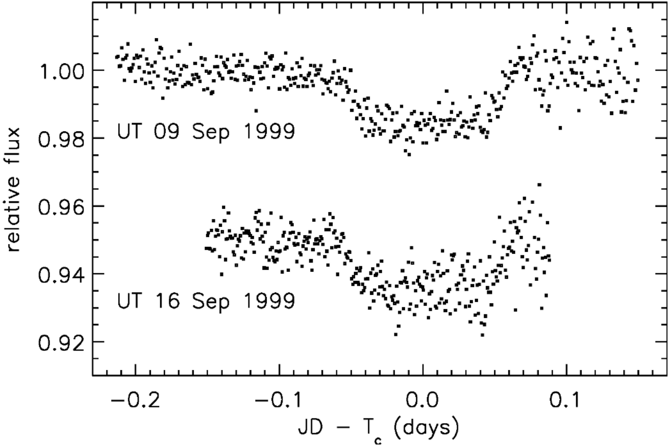
1. Historical Context



HD 209458 b first transit, STARE 1999. Charbonneau et al. (2000)

From Zero to Six Thousand in Three Decades

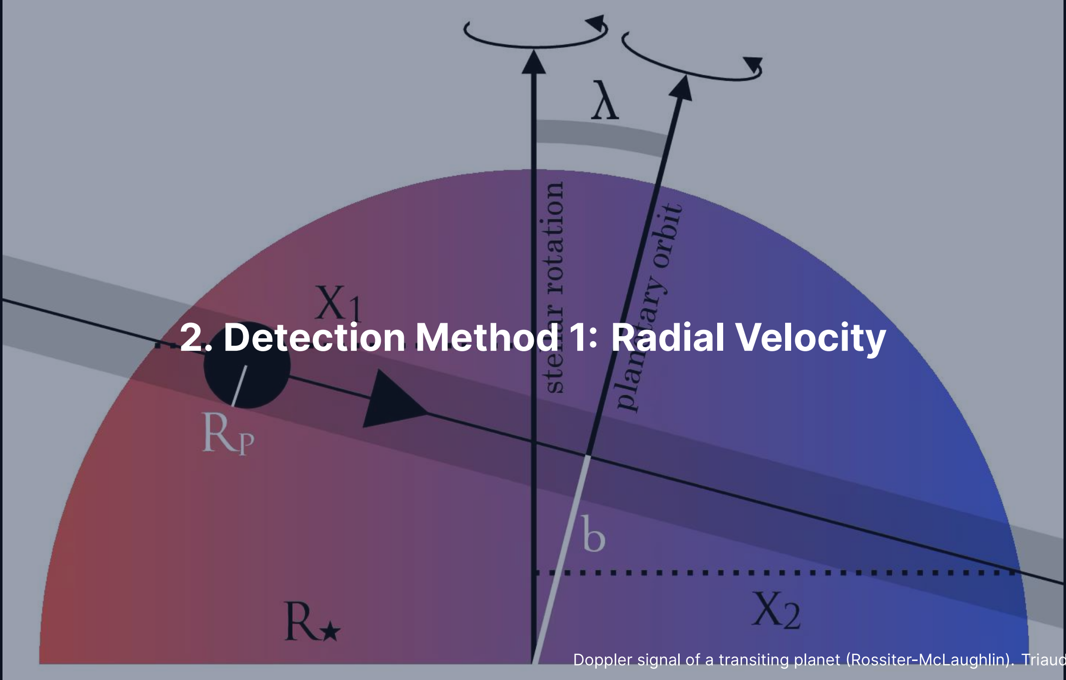
- **1992 (PSR B1257+12):** first exoplanets, around a pulsar, by timing (Wolszczan & Frail)
- **1995 (51 Pegasi b):** first planet around a Sun-like star; a 4.23-day orbit forced migration theory (Mayor & Queloz)
- **1999 (HD 209458 b):** first transit of an RV planet: a gas-giant radius and bulk density



The first ground-based exoplanet transit: HD 209458 b with the STARE photometer on two nights in September 1999. Charbonneau et al. (2000).

Successive transits dropped the flux by about 1.7% (independent detection by Henry et al. 2000): exoplanets became physical objects with measurable radii.

2. Detection Method 1: Radial Velocity



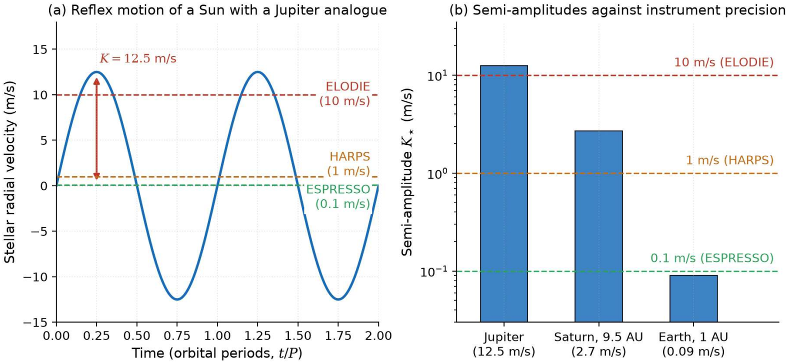
Doppler signal of a transiting planet (Rossiter-McLaughlin). Triaud (2018)

Radial Velocity: Doppler Reflex of the Star

- Star and planet orbit a common barycentre: $a_\star = (m_p / M_\star)\, a_p$
- The periodic Doppler shift of the stellar absorption lines gives the reflex velocity
- Semi-amplitude $K_\star$ for inclination i (edge-on: $\sin i = 1$; circular: $e = 0$):

$$K_\star = \left(\frac{2\pi G}{P}\right)^{1/3} \frac{m_p \sin i}{M_\star^{2/3}} \frac{1}{\sqrt{1 - e^2}}$$

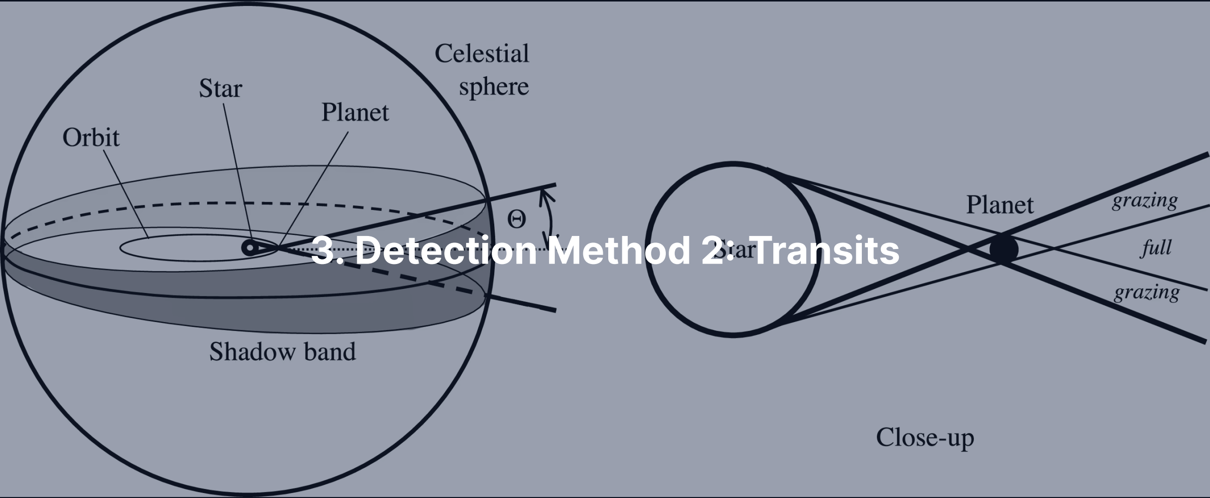
Jupiter at 5.2 AU: $K_\star \approx 12.5\ \mathrm{m\ s^{-1}}$; Earth at 1 AU: $K_\star \approx 9\ \mathrm{cm\ s^{-1}}$.



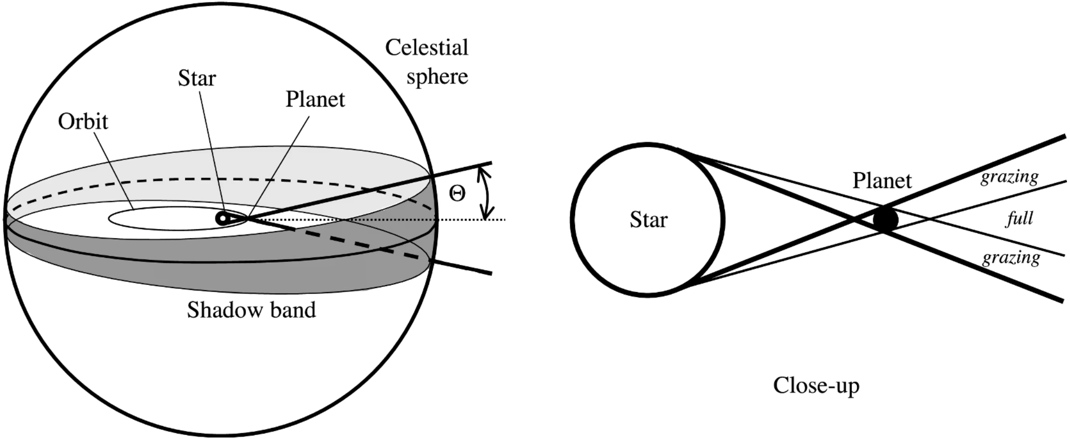
(a) The reflex velocity of a Sun with a Jupiter analogue against the precisions of ELODIE, HARPS and ESPRESSO; (b) the semi-amplitudes of Jupiter, Saturn and Earth analogues. Course figure.

An Earth analogue at 0.09 m s^{-1} sits below the instrumental floor and within the 0.1–1 m s^{-1} stellar jitter; RV alone gives only the minimum mass $m_p \sin i$.

3. Detection Method 2: Transits



Transit geometry. Winn (2010)



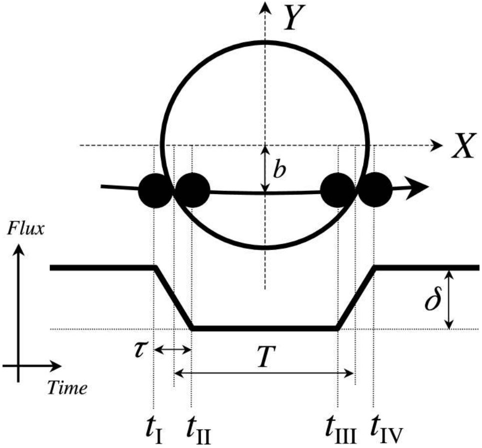
Transit geometry: a planet whose orbital plane lies near the line of sight crosses the stellar disk. Winn (2010).

Transit depth $\delta = (R_p/R_\star)^2$ and geometric probability $p \approx R_\star/a$: a strong selection bias toward short periods and large $R_p/R_\star$.

Quick Scales

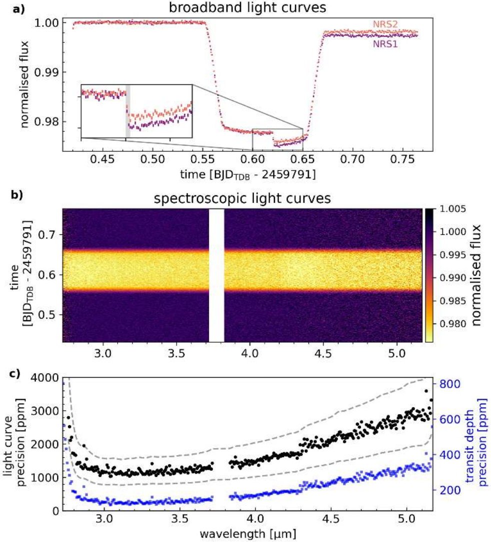
Transit depth scales inversely with the square of the stellar radius, so small planets around small stars are far easier to detect.

- **Jupiter around Sun:** $\delta \approx 1.01\%$, easily detected from ground-based telescopes
- **Earth around Sun:** $\delta \approx 84$ ppm, requires ultra-precise space photometry
- **Earth around M dwarf ($0.15\,R_{\odot}$):** $\delta \approx 4000$ ppm, accessible from ground and space



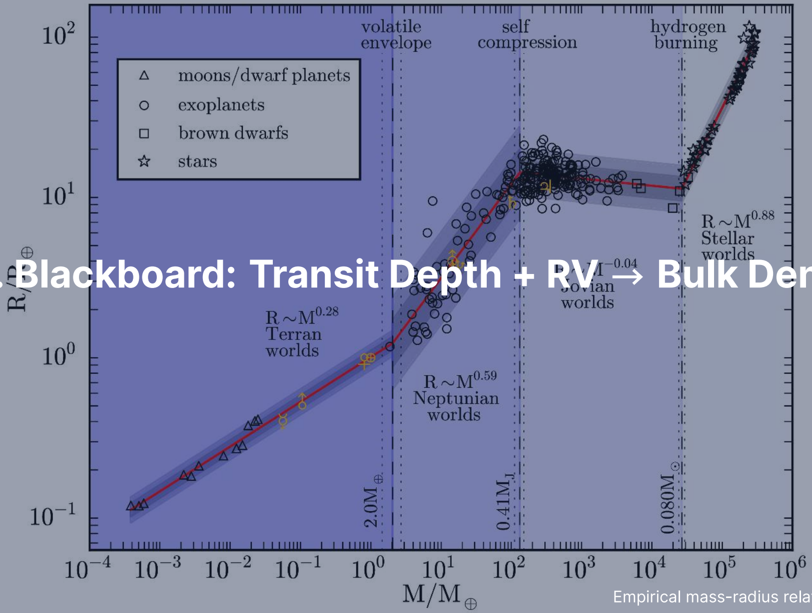
Anatomy of a transit light curve with its four contact times t_I to t_IV. Winn (2010).

The depth gives the radius ratio $R_p/R_\star$; the total duration and the ingress time give the impact parameter b and the inclination. A real transit bottom curves from stellar limb darkening.



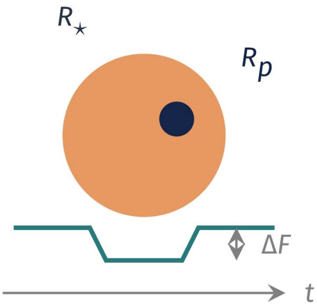
JWST/NIRSpec transit photometry of WASP-39 b: depths measured to 200–600 ppm per spectroscopic bin, below typical exoplanet absorption features. Alderson et al. (2023).

4. Blackboard: Transit Depth + RV → Bulk Density



Empirical mass-radius relation. Chen & Kipping (2017)

Goal and strategy



Goal: a radius and a mass for the same planet, hence a density.

- Transit alone: the radius R_p, but no mass
- RV alone: the minimum mass $m_p \sin i$, but no radius
- Combined: **both**, so the bulk density $\bar{\rho}_p$

Bulk density turns an astronomical detection into a physical, geological body. To the board.

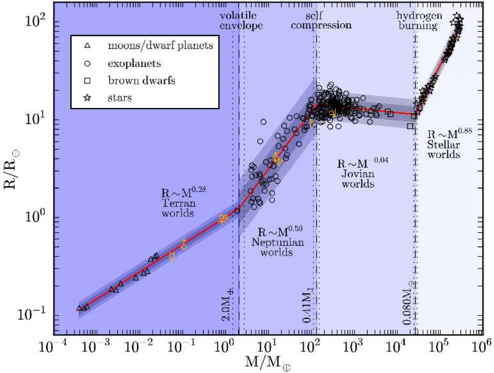
Result: Depth, Semi-Amplitude, Density

Transit depth $\delta = (R_p/R_\star)^2$ gives R_p; the reflex velocity $K_\star = (2\pi G/P)^{1/3}\, m_p \sin i\, M_\star^{-2/3}$ gives $m_p \sin i$ (board).

- A transit forces $\sin i \approx 1$ and collapses the $m_p \sin i$ degeneracy
- Both R_p and m_p measured for the same object
- Bulk density follows from elementary geometry:

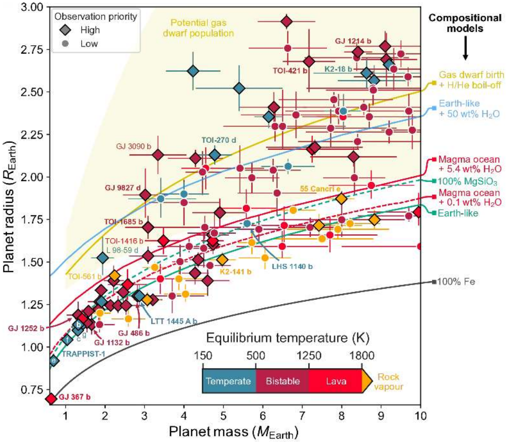
$$\boxed{\bar{\rho}_p = \frac{m_p}{(4/3)\pi R_p^3} = \frac{3m_p}{4\pi R_p^3}}$$

Transit + RV is the single observational machinery that took exoplanet science from exotic claim to compositional census.



The empirical mass-radius relation of exoplanets and solar-system bodies. Chen & Kipping (2017).

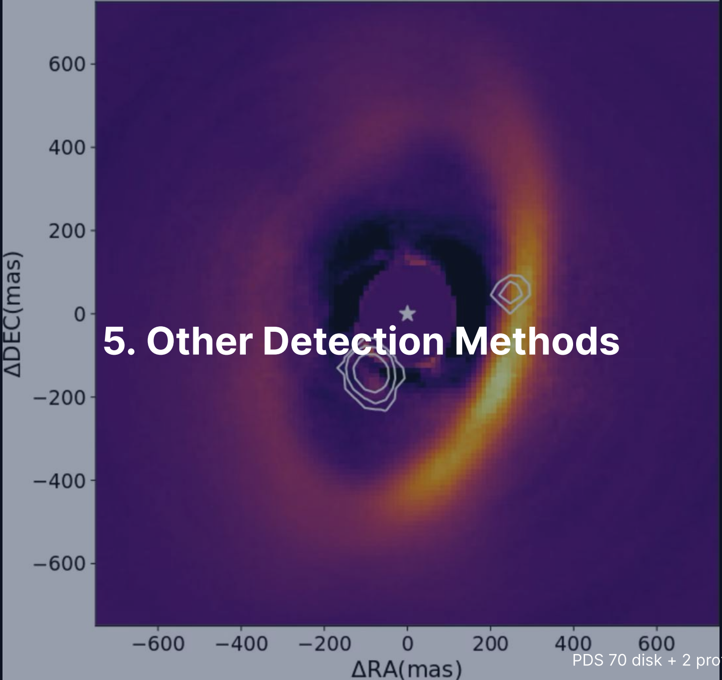
Density maps to composition: $\bar{\rho}$ ≈ 5.5 $\mathrm{g\,cm^{-3}}$ rocky (Earth, Venus), 1.3 gas giant (Jupiter), 0.5 an inflated envelope, and 2–4 $\mathrm{g\,cm^{-3}}$ the sub-Neptunes unseen in the solar system.



Mass against radius for small exoplanets with interior models from pure iron to volatile-rich envelopes, sorted by thermal regime. Reproduced from Lichtenberg et al. (2025).

Rocky super-Earths follow the silicate-iron curves; sub-Neptunes need substantial volatile envelopes or water-rich layers to explain their larger radii.

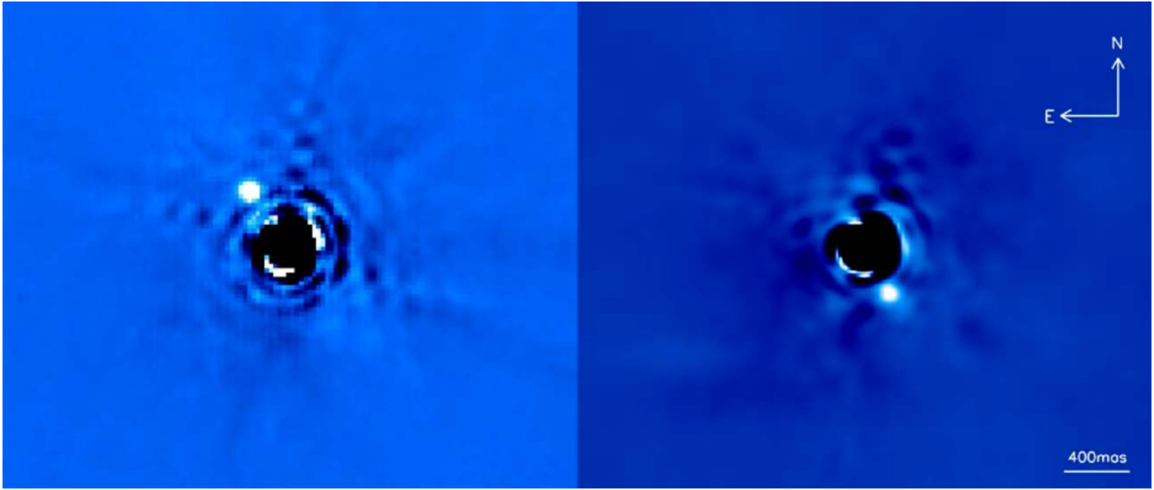
5. Other Detection Methods



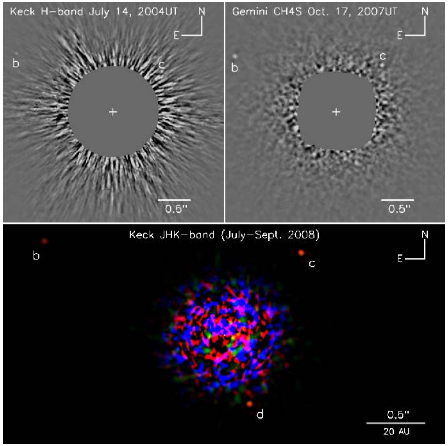
PDS 70 disk + 2 protoplanets. Haffert et al. (2019)

Direct Imaging: Spatial Separation of Photons

- **Contrast:** suppress the star by 10^{-9} (Jupiter) to 10^{-10} (Earth twin) at separation $\theta = a/d$
- **Technology:** extreme adaptive optics plus coronagraphs against diffraction and speckle noise
- **Bias:** young, massive, self-luminous giants on wide orbits (a > 10 AU)

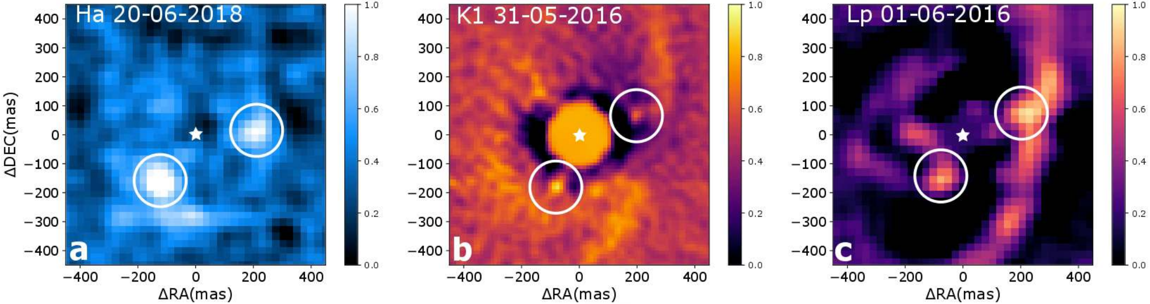


Direct image of β Pictoris b: orbital motion between 2003 and 2009 confirms a bound ~ $9\,M_{\mathrm{J}}$ companion at ~ 9 AU. Lagrange et al. (2010); credit: ESO/A.-M. Lagrange et al.

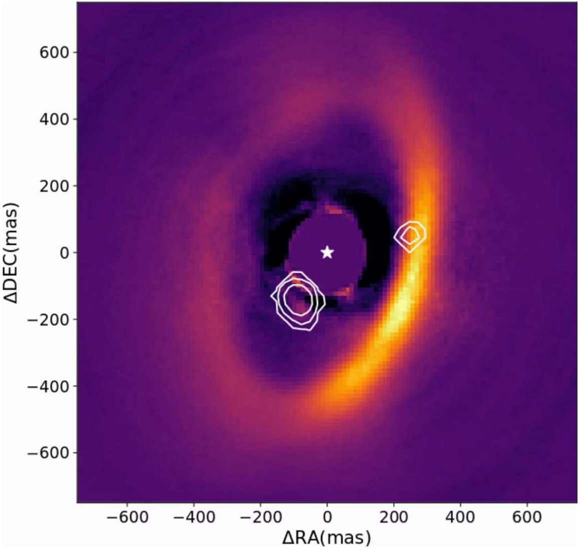


Keck and Gemini adaptive-optics discovery images of the HR 8799 planets. Marois et al. (2008).

Four co-moving planets of 5–10 M_J at 14–70 AU (planet e added in 2010) around a 30–60 Myr star: they still glow from their contraction heat.



PDS 70 b and c inside the disk gap in MUSE Hα (a), SPHERE K1 (b) and NACO L′ (c): both protoplanets are accreting within the cleared cavity. Haffert et al. (2019).



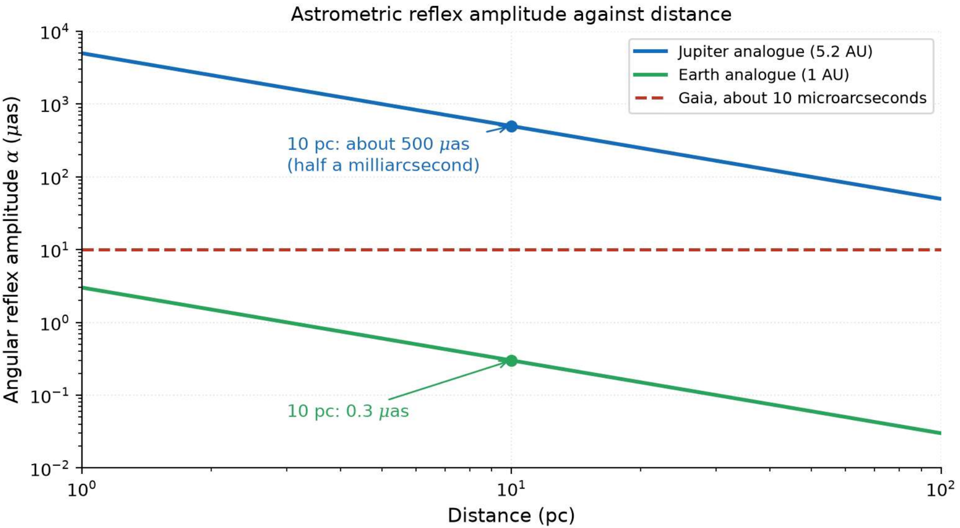
Composite scattered-light image of the PDS 70 protoplanetary disk with the two embedded protoplanets inside the cleared gap. Credit: Haffert et al. (2019).

Astrometry: Gaia Will Find Jupiter Analogues

Astrometry follows the star's reflex wobble on the sky and gives true masses.

$$\alpha = \frac{m_p}{M_\star} \cdot \frac{a_p}{d}$$

- **Signal:** a Jupiter analogue at 10 pc gives $\alpha \sim 500\,\mu$as; an Earth analogue $\sim 0.3\,\mu$as
- **Gaia DR4 (2026):** thousands of cold giants expected; $\sim 10\,\mu$as final precision
- **True mass:** the inclination i is measured, so no $m \sin i$ degeneracy

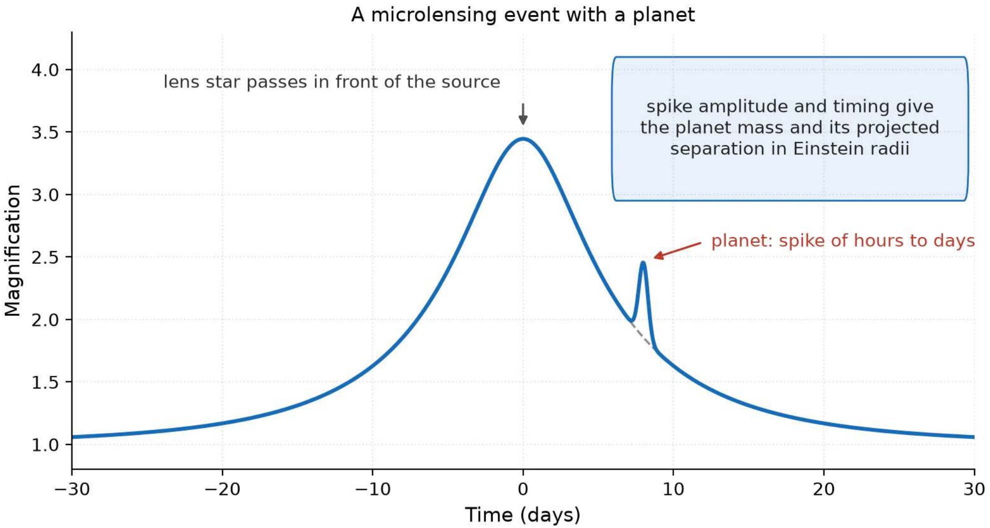


Astrometric reflex amplitude against distance for Jupiter and Earth analogues, with Gaia's final precision of about 10 microarcseconds as the dashed line. Course figure.

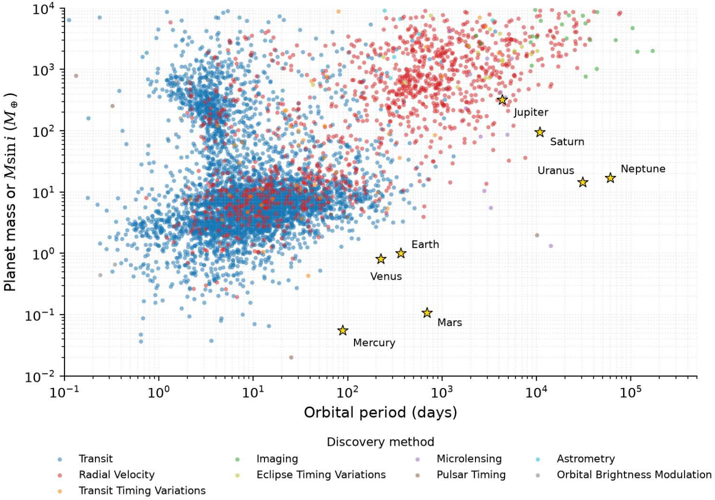
Microlensing, Timing, and Detection Biases

- **Microlensing (Roman):** cool planets at 1–10 AU and unbound rogue worlds
- **Timing variations (TTVs):** dynamical masses for faint hosts such as TRAPPIST-1
- **Together:** transits and RV probe close-in orbits; imaging, astrometry and microlensing the wide ones

Each method covers its own part of parameter space, so the archive carries the biases of the methods.

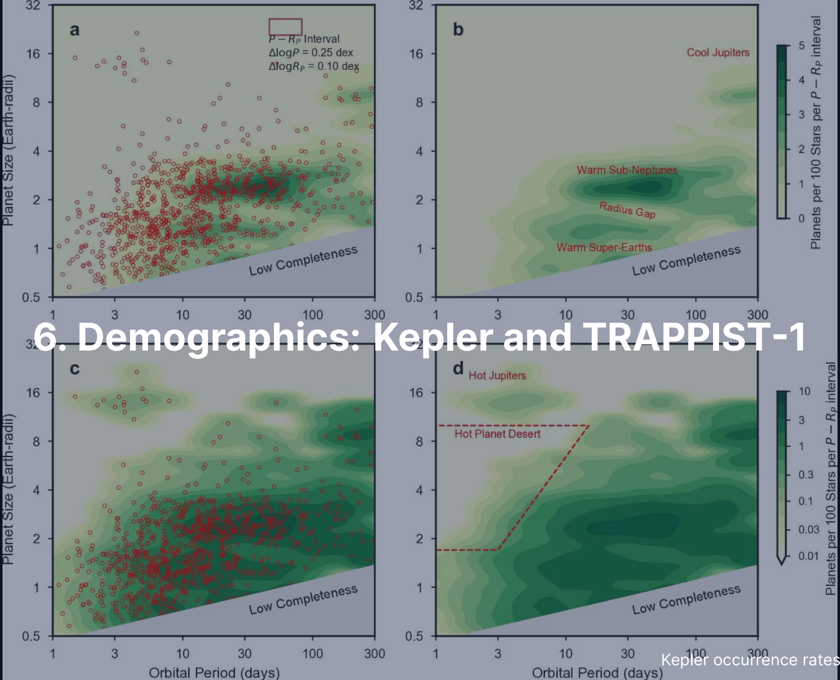


A microlensing event with a planet: the magnification of a background source by a lens star, and the spike of hours to days that a planet around the lens adds. Course figure.



Confirmed exoplanet masses (or $m_p \sin i$) against orbital period by detection technique, with the solar-system planets: discoveries pile up at high mass and short period. Credit: NASA Exoplanet Archive (2026). Course figure.

6. Demographics: Kepler and TRAPPIST-1

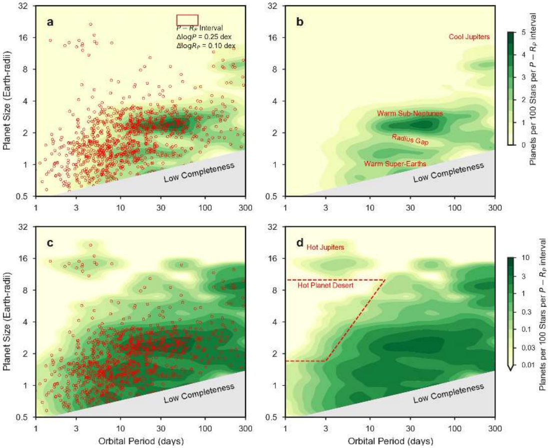


Kepler occurrence rates. Petigura et al. (2018)

The Kepler Revolution

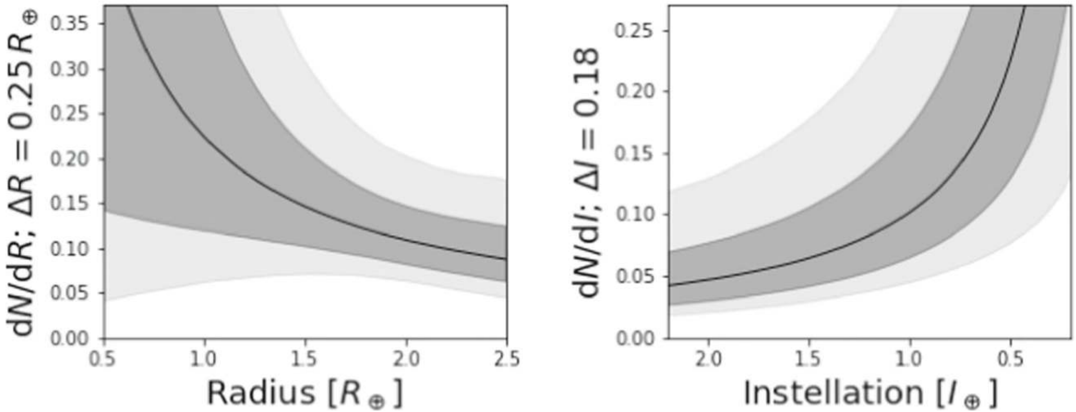
- **Planets are ubiquitous:** on average more than 1 planet per main-sequence star (Petigura et al. 2018)
- **Small planets dominate:** super-Earths and sub-Neptunes are common; hot Jupiters are rare (0.5–1%)
- **Habitable terrestrial rate:** Earth-sized planets in the habitable zone occur at $\eta_\oplus \sim 0.4$ (Bryson et al. 2021)

Kepler's computable detection efficiency gave the first bias-corrected occurrence rates: planets outnumber stars in the Milky Way.



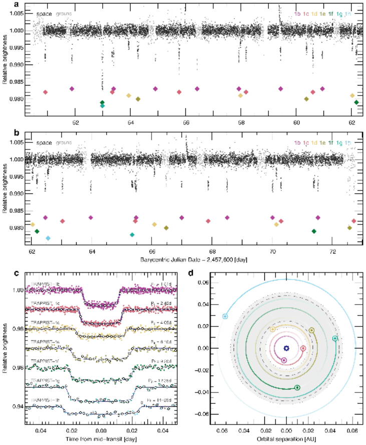
Planet occurrence against orbital period and size from the California-Kepler Survey after stellar parameter refinement. Petigura et al. (2018).

Small planets outnumber giants at every period; the fall-off beyond ~ 100 days is partly completeness-limited.



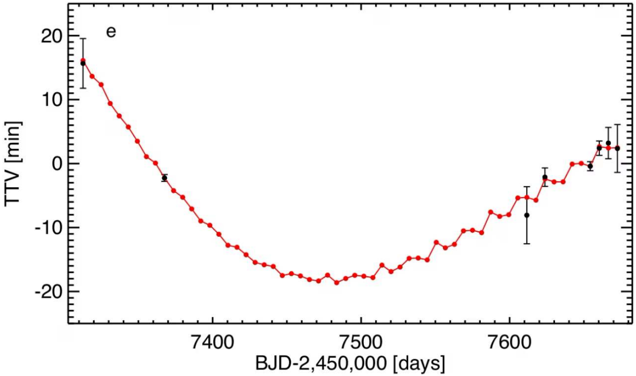
The occurrence rate of Earth-sized planets in the habitable zones of Sun-like stars, marginalised over radius and instellation, with 68% and 95% credible bands. Bryson et al. (2021).

The central estimate $\eta_\oplus \sim 0.4$: terrestrial planets in habitable zones are common.



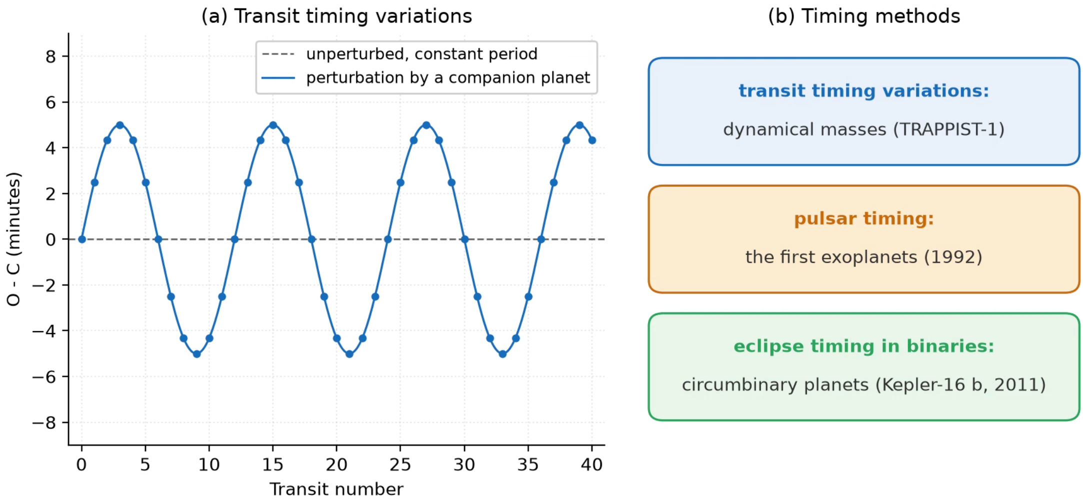
Transits of the seven TRAPPIST-1 planets. Gillon et al. (2017).

An M8 ultra-cool dwarf at 12 pc with seven Earth-sized planets within 0.06 AU in a Laplace-like resonant chain, consistent with early inward migration; e, f and g orbit in the temperate habitable zone.



Transit-timing variations of TRAPPIST-1 e: observed deviations against a dynamical model coupled to the other six planets. Gillon et al. (2017).

Timing variations of tens of minutes, fitted with simultaneous N-body models, break the mass degeneracies and give individual masses and densities without radial velocities.

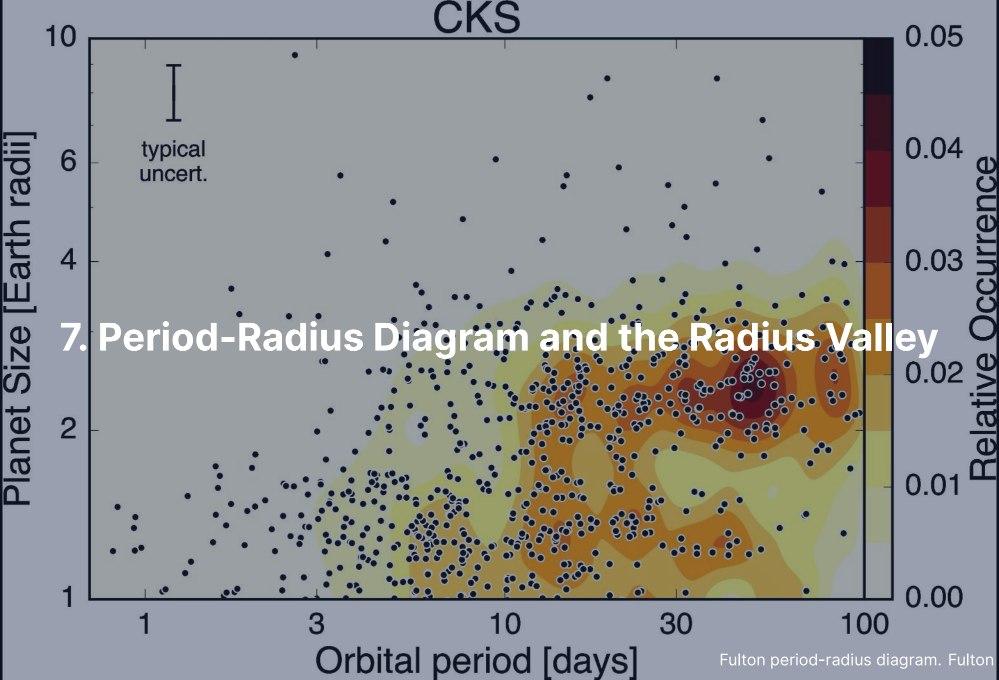


(a) Observed minus calculated transit times of a planet perturbed by a companion (5 minute amplitude, schematic); (b) the three timing methods: transit timing, pulsar timing and eclipse timing in binaries. Course figure.

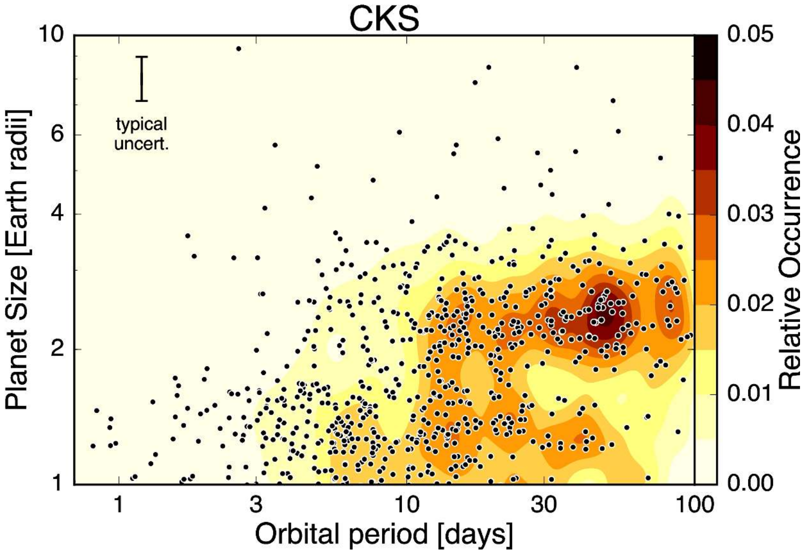
Break

Earthrise, Apollo 8 (1968). Credit: NASA

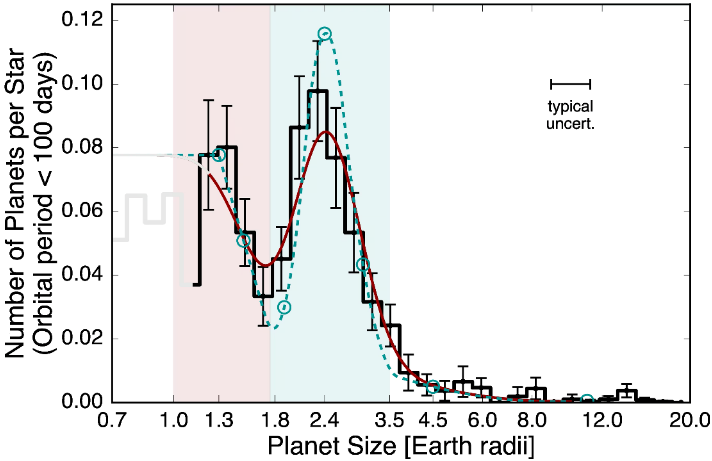
7. Period-Radius Diagram and the Radius Valley



Fulton period-radius diagram. Fulton et al. (2017)



Period-radius distribution of small Kepler planets after stellar parameter refinement: super-Earths and sub-Neptunes separated by a deficit at ~ 1.8 $R_{\oplus}$. Fulton et al. (2017).



The radius valley: the size distribution of small Kepler planets is bimodal. Fulton et al. (2017).

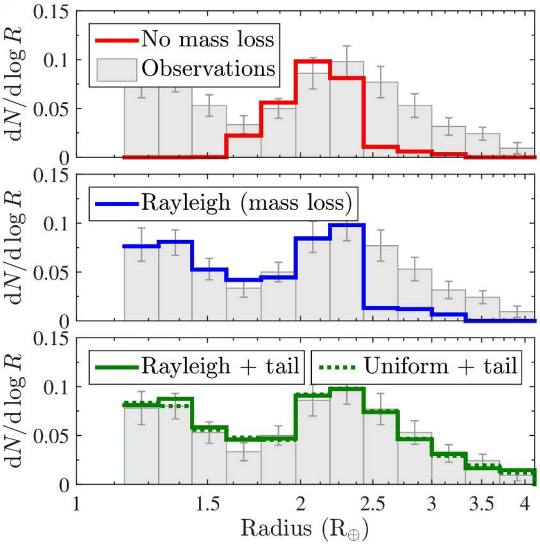
A deficit at 1.5–2.0 $R_\oplus$ separates super-Earths near 1.3 $R_\oplus$ (bare rocky cores) from sub-Neptunes near 2.4 $R_\oplus$ (volatile envelopes); photoevaporation or core-powered mass loss carves it.

Photoevaporation Strips H/He Envelopes

Energy-limited hydrodynamic escape driven by high stellar X-ray and EUV irradiation:

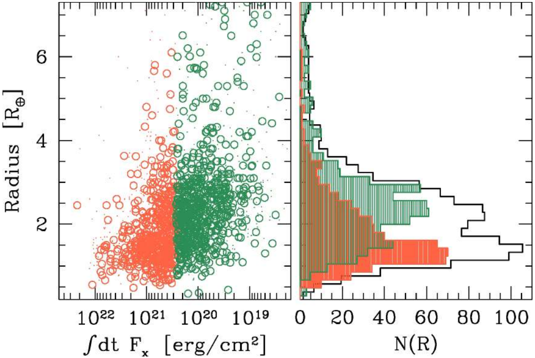
$$\dot{M} \approx \frac{\epsilon \pi F_{\mathrm{XUV}} R_p^3}{G M_p}$$

- Young stars saturate XUV flux at ~ $10^3\times$ modern levels for ~ 100 Myr
- Strips light primordial H/He envelopes from lower-mass cores at close-in orbits
- Owen & Wu (2013): naturally carves the bimodal radius valley

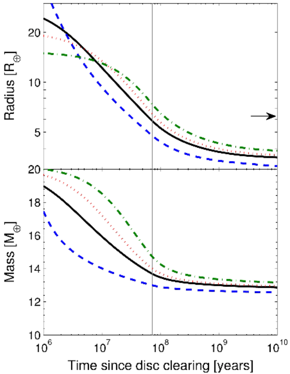


Core-powered mass loss: envelope loss driven by heat from the cooling rocky interior also produces a bimodal distribution with a gap at ~ 1.8 $R_\oplus$. Credit: Ginzburg et al. (2018).

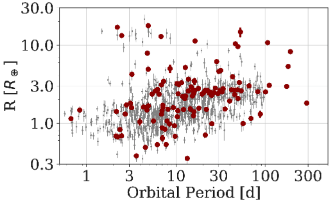
Photoevaporation Model + Mass-Loss Tracks

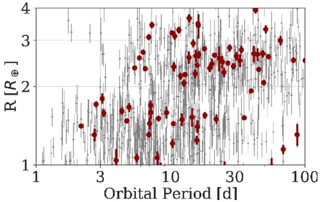


Left: predicted radius valley



Right: time evolution of radius and mass; the saturated XUV phase drives the bimodal outcome (Owen & Wu 2013).

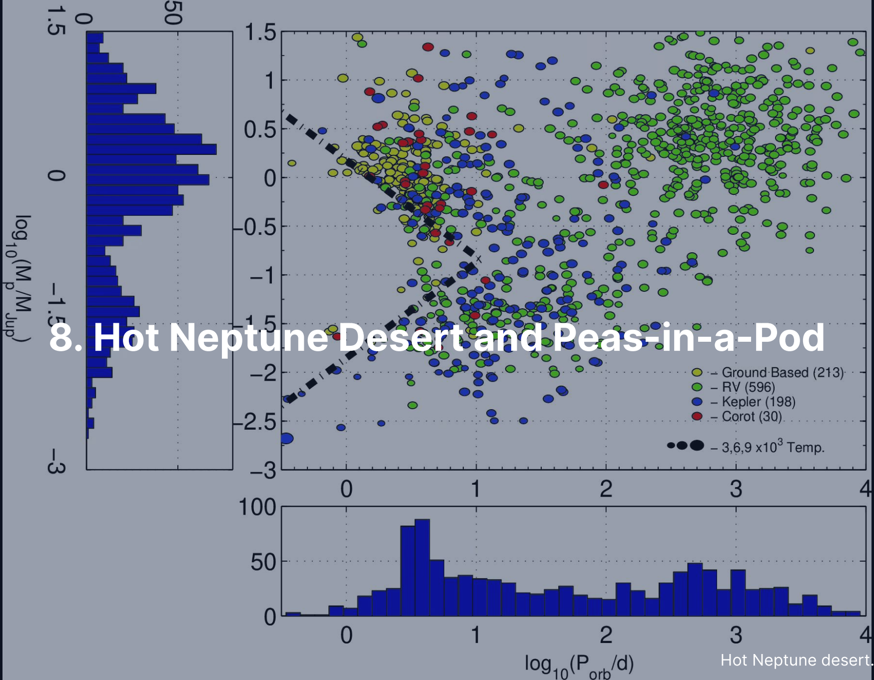




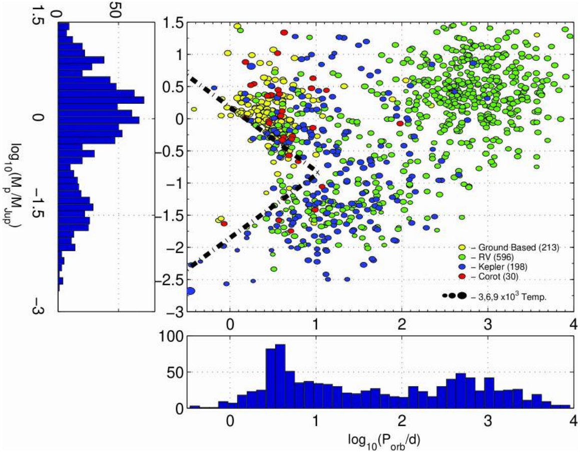
The radius valley with asteroseismic stellar radii: its position shifts with orbital period. Van Eylen et al. (2018).

The valley slopes to smaller R_p at longer periods, as photoevaporation and core-powered cooling predict: many close-in super-Earths are stripped cores rather than primordial rocky worlds.

8. Hot Neptune Desert and Peas-in-a-Pod



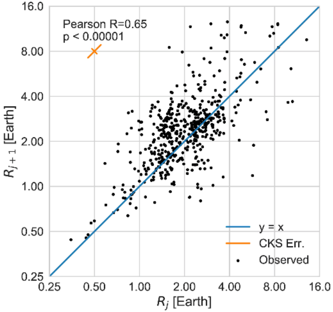
Hot Neptune desert. Mazeh et al. (2016)



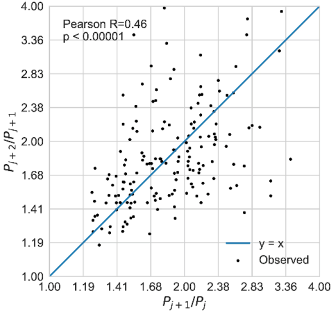
The hot Neptune desert in the mass-period plane. Mazeh et al. (2016).

A deficit of 10–100 $M_\oplus$ planets at $P < 5$ d: at the **bottom edge** photoevaporation strips low-mass envelopes and leaves super-Earths, at the **top edge** tidal Roche-lobe overflow strips inflated giants.

Peas-in-a-Pod Architectures



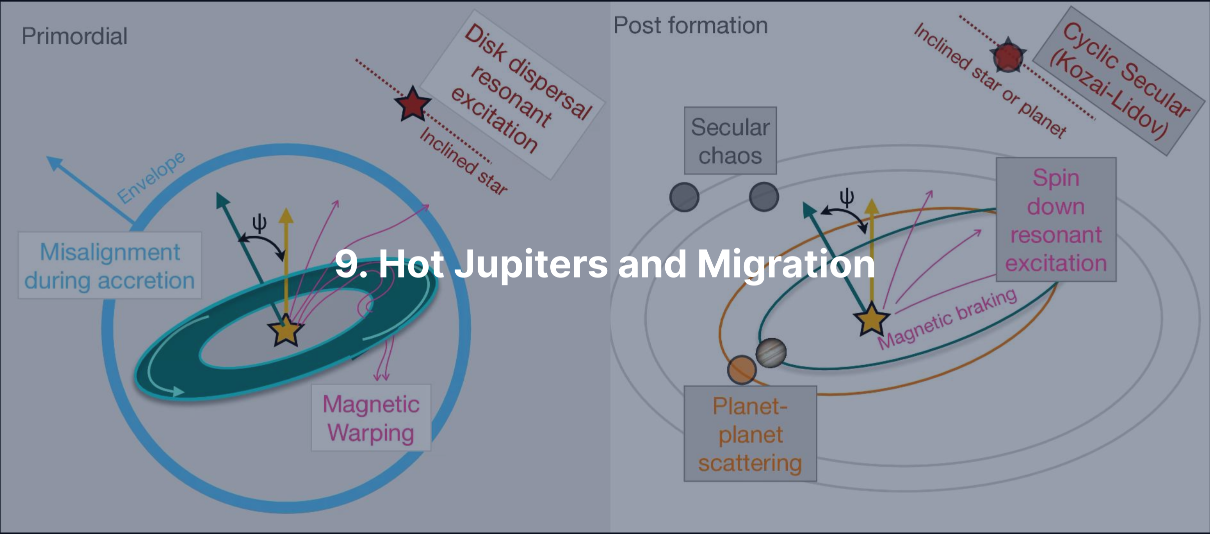
Adjacent radii: Pearson r = 0.65



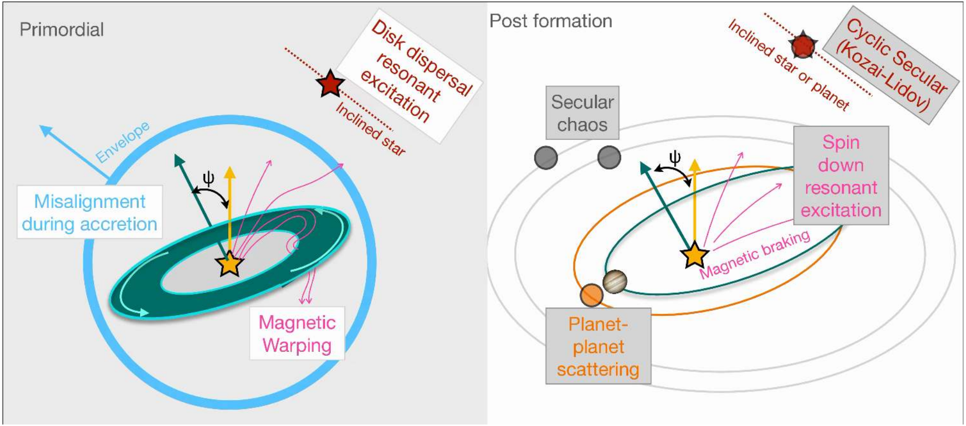
Adjacent period ratios: clustered

- Planets in compact multi-systems tend to have similar sizes and uniform orbital spacing (Weiss et al. 2018)
- Favours quiet in-situ growth or migration over chaotic giant impacts

9. Hot Jupiters and Migration

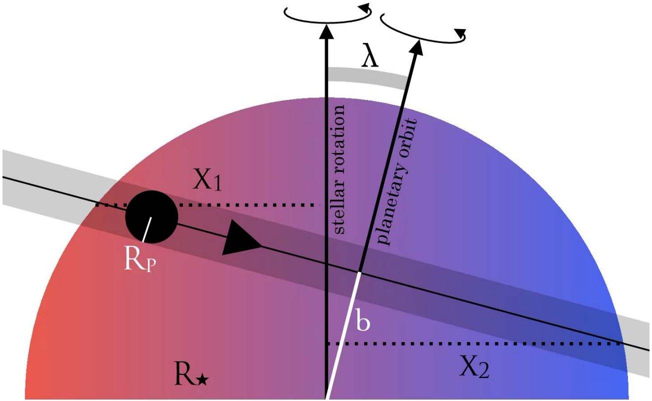


Hot Jupiter migration channels. Albrecht et al. (2022)



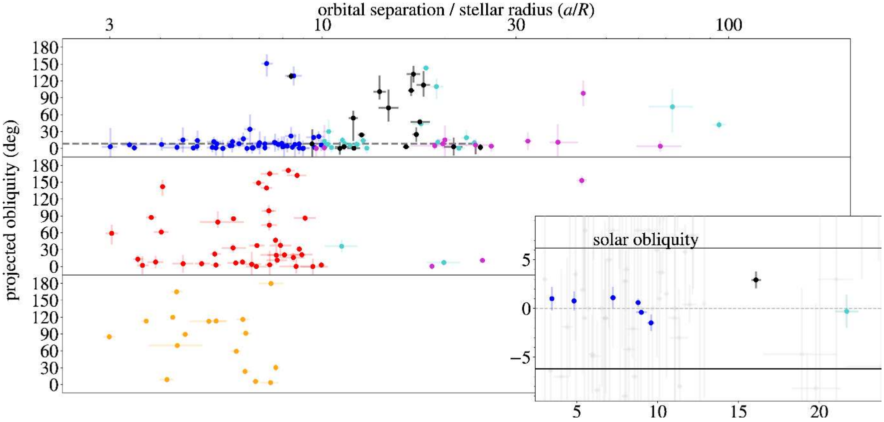
Three pathways to a hot Jupiter and the stellar obliquity each one leaves. Albrecht et al. (2022).

Disk migration (Type II) keeps a low obliquity; **high-eccentricity** Kozai-Lidov migration and **planet scattering** leave wide misalignments; in situ growth at 0.05 AU fails for lack of solid mass.



The Rossiter-McLaughlin effect: the Doppler signature of a planet crossing a rotating star. Triaud (2018).

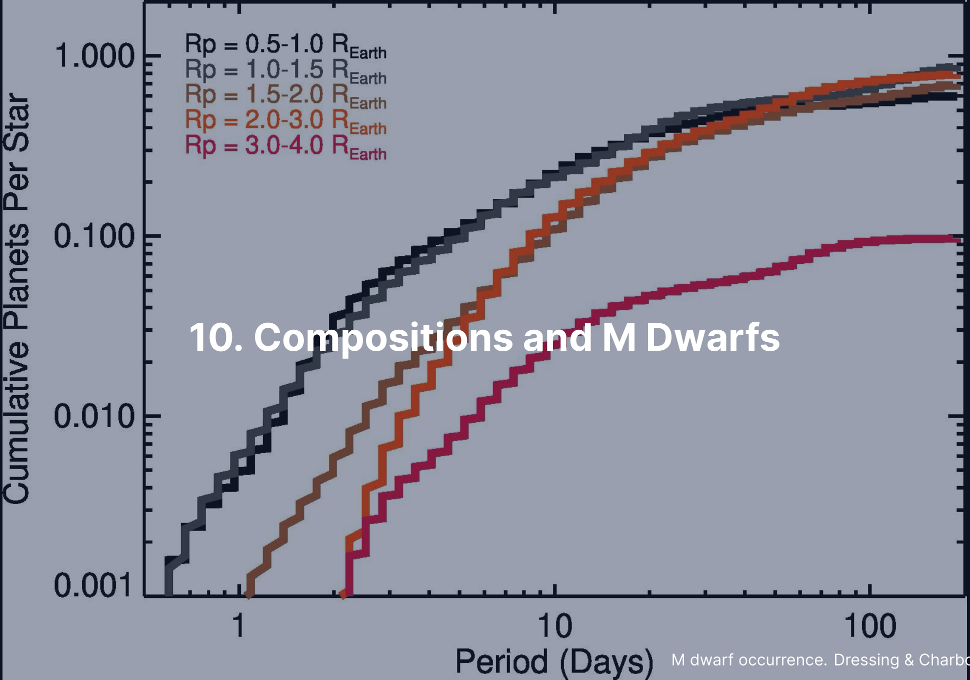
On an aligned orbit the planet first hides the approaching, then the receding hemisphere; the time-resolved line distortion gives the sky-projected spin-orbit angle λ, which separates disk migration from dynamical scattering.



Observed stellar obliquities of hot Jupiter hosts against $a/R_\star$, grouped by stellar effective temperature. Albrecht et al. (2022).

Cool stars ($T_{\rm eff}$ < 6250 K) realign tidally through their convective envelopes; **hot stars** keep high obliquities; the wide dispersion at large $a/R_\star$ shows that dynamical migration occurs.

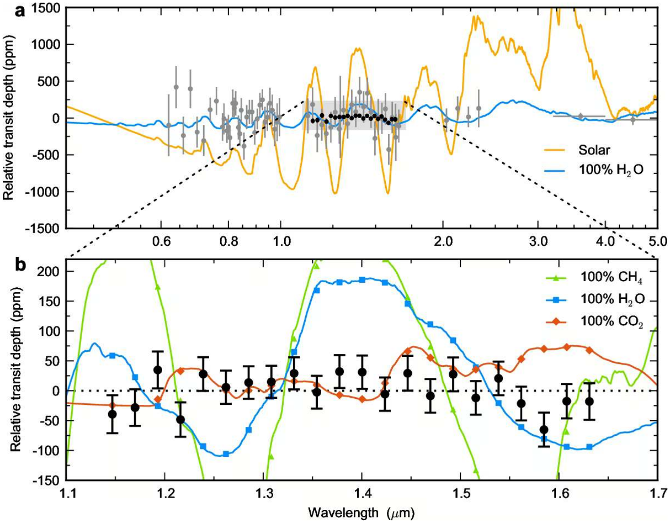
10. Compositions and M Dwarfs



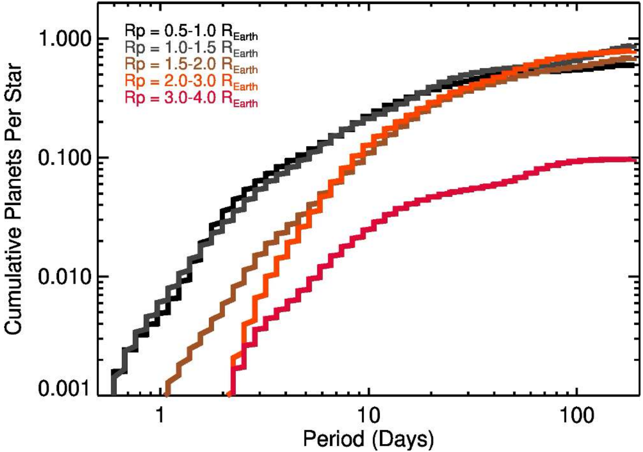
M dwarf occurrence. Dressing & Charbonneau (2015)

Sub-Neptunes: Water World or H/He?

- **Density split:** super-Earths ($\bar{\rho} \sim 4$–$8\ \mathrm{g\,cm^{-3}}$) are rocky; sub-Neptunes ($\bar{\rho} \sim 1$–$3\ \mathrm{g\,cm^{-3}}$) require volatile envelopes
- **Degeneracy:** density cannot separate a 1% H/He envelope from a 50% water-rich mantle
- **Spectroscopic test:** transmission spectroscopy gives the atmospheric mean molecular weight



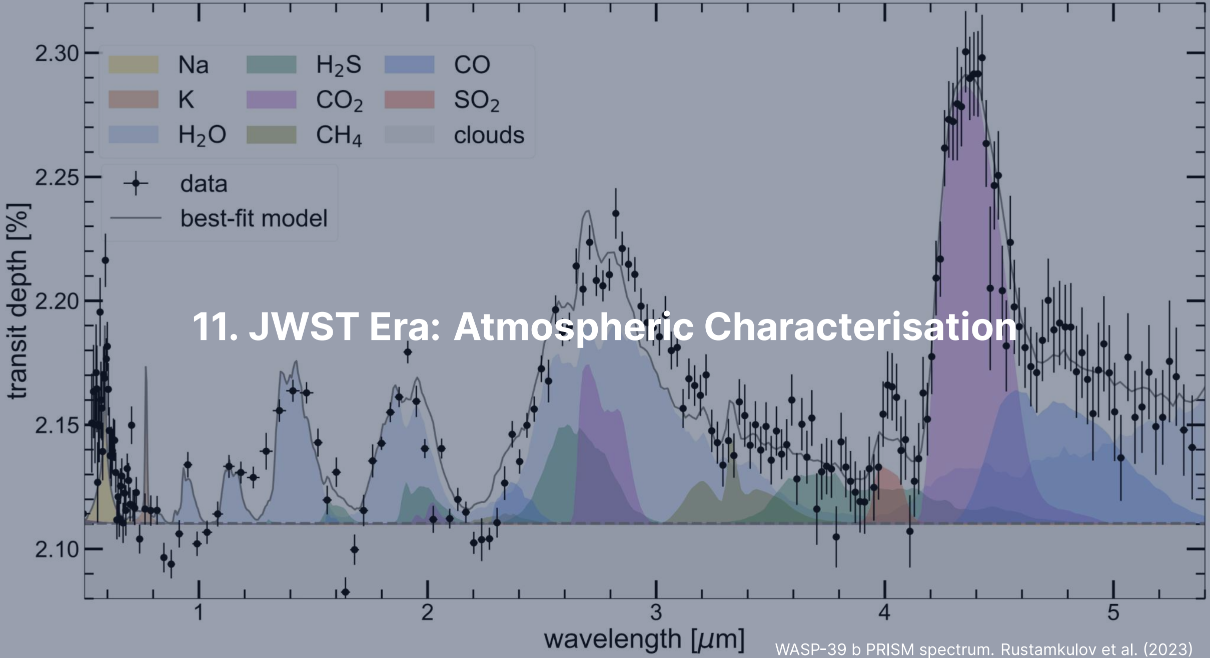
Featureless HST transmission spectrum of the warm sub-Neptune GJ 1214 b: cloud-free $\mathrm{H_2O}$, $\mathrm{CH_4}$ and $\mathrm{CO_2}$ atmospheres are ruled out, so high clouds or hazes are present. Credit: Kreidberg et al. (2014).



Planet occurrence around M dwarfs from Kepler. Dressing & Charbonneau (2015).

M dwarfs host ~ 2.5 small planets each and favour transits and RV, but **XUV desiccation** in the long pre-main sequence can strip whole oceans (Luger & Barnes 2015) and **tidal locking** fixes day and night sides.

11. JWST Era: Atmospheric Characterisation



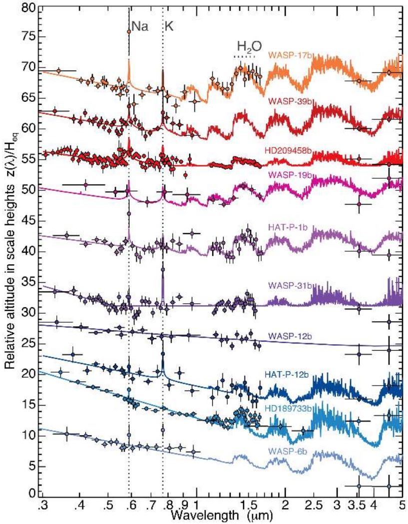
WASP-39 b PRISM spectrum. Rustamkulov et al. (2023)

Transmission Spectroscopy

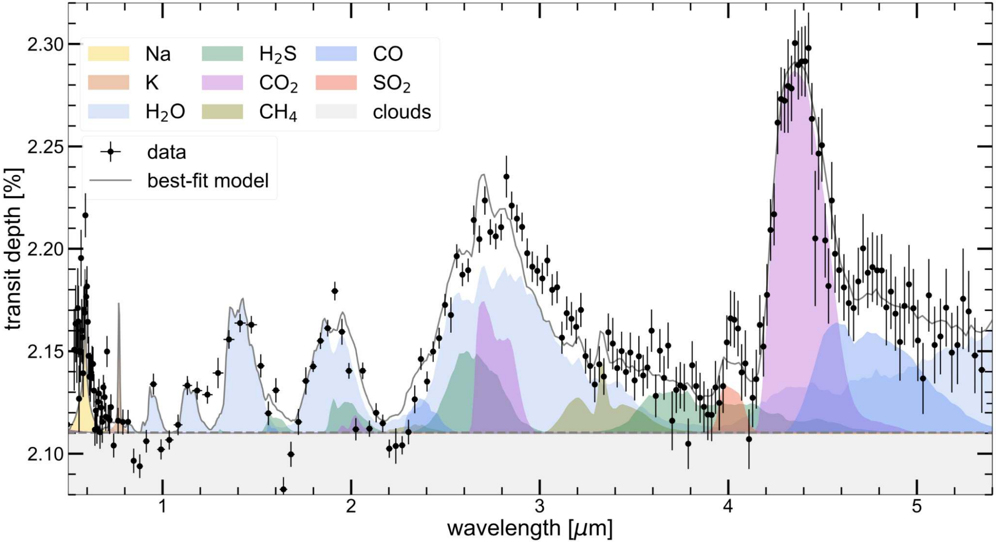
Starlight filtered through an atmosphere during transit encodes molecular absorption; with the scale height

$$H = \frac{k_{\mathrm{B}} T}{\mu m_u g}, \qquad \delta(\lambda) \approx \frac{(R_p + n_H(\lambda)\, H)^2}{R_\star^2}, \qquad \frac{\Delta\delta}{\delta} \approx \frac{2 n_H H}{R_p}.$$

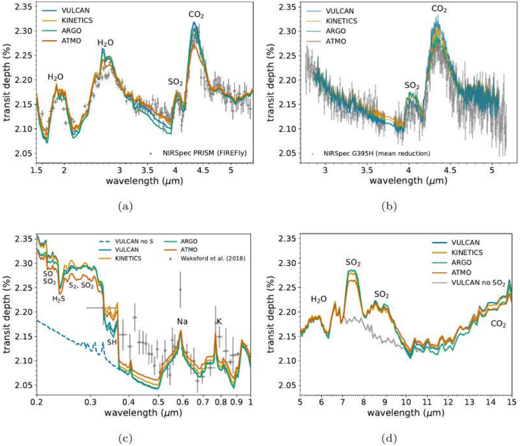
- Hot Jupiter (T = 1500 K, μ = 2.3, g = 25 $\mathrm{m\,s^{-2}}$): $H \approx$ 200 km, $\Delta\delta/\delta \approx$ 2%, absolute modulation $\Delta\delta \sim$ 200 ppm (JWST-routine)
- Sub-Neptunes: 10–100 ppm; terrestrial atmospheres: < 10 ppm
- High clouds or hazes flatten the spectrum (GJ 1214 b): a flat spectrum does not mean no atmosphere



Transmission spectra of ten hot Jupiters from HST and Spitzer across 0.3–5 $\mu\mathrm{m}$: a continuum from clear atmospheres with water absorption to cloudy scattering slopes. Credit: Sing et al. (2016).

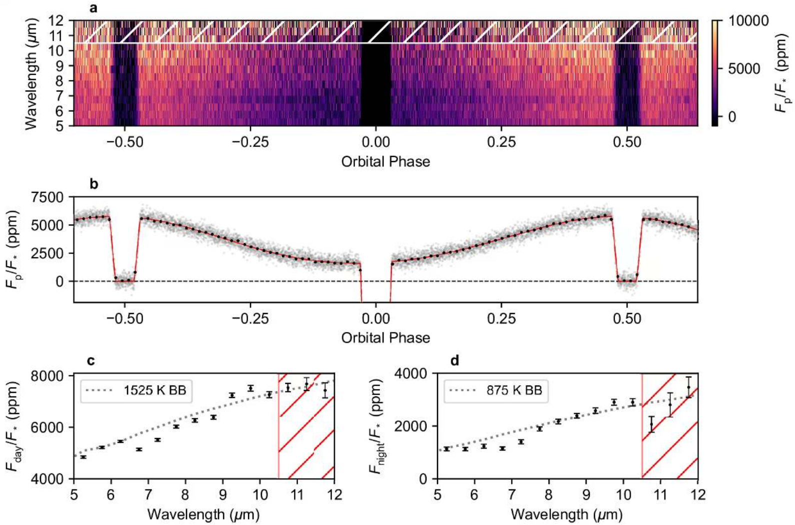


JWST/NIRSpec PRISM transmission spectrum of WASP-39 b with model opacity bands: $\mathrm{H_2O}$ (33σ), $\mathrm{CO_2}$ (28σ), clouds (21σ), Na (19σ) and CO (7σ). Rustamkulov et al. (2023).



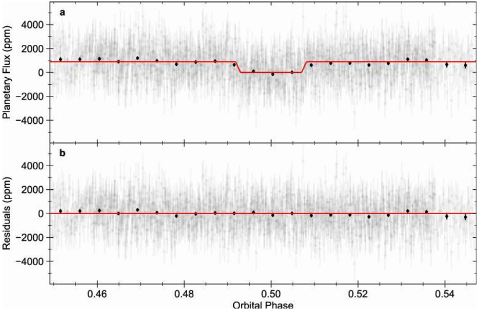
The 4.05 $\mu\mathrm{m}$ absorption feature of $\mathrm{SO_2}$ in the JWST spectrum of the hot Saturn WASP-39 b ($0.28\,M_{\mathrm{J}}$). Tsai et al. (2023).

Photochemical $\mathrm{SO_2}$ from UV photolysis of $\mathrm{H_2S}$: the first photochemical product identified in an exoplanet atmosphere, proof of disequilibrium chemistry.

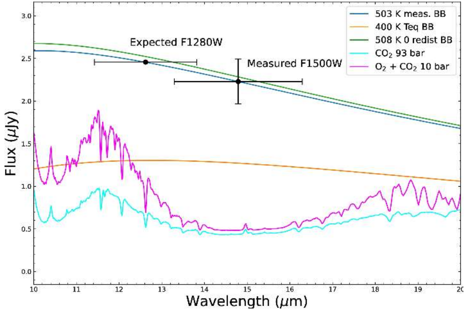


JWST/MIRI spectroscopic phase curve of WASP-43 b: thermal emission from dayside to nightside constrains the atmospheric heat redistribution. Credit: Bell et al. (2024).

TRAPPIST-1 b: Bare Rock Dayside

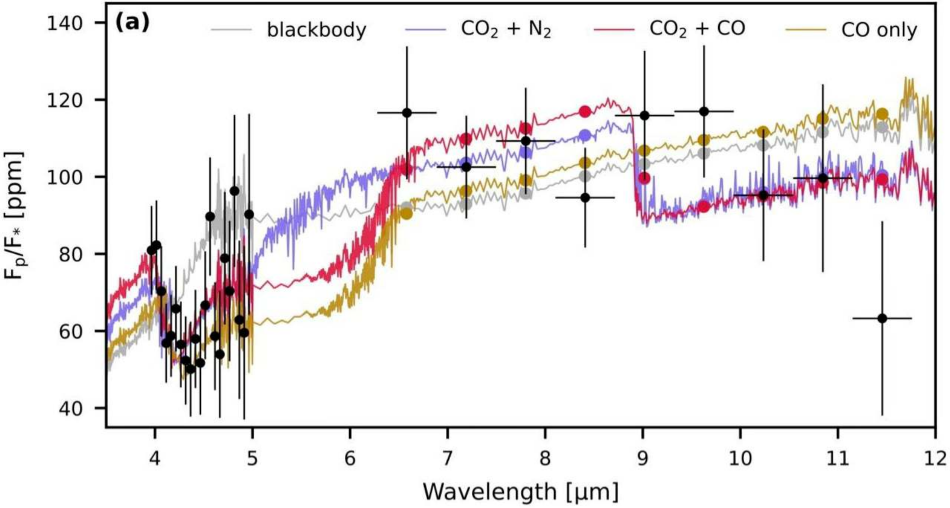


15 $\mu\mathrm{m}$ secondary eclipse

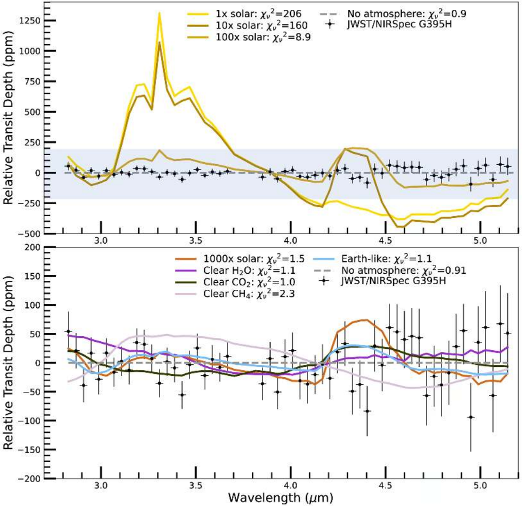


Compared to model atmospheres

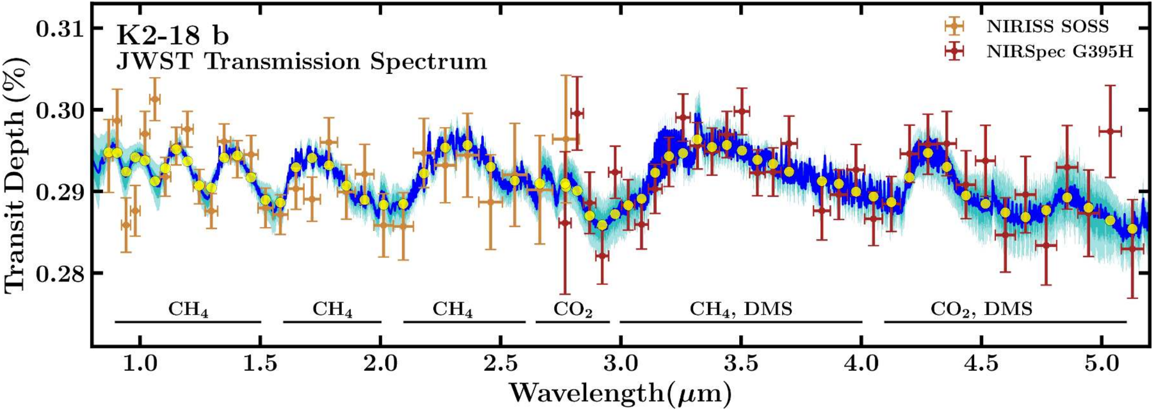
$T_B = 503^{+26}_{-27}$ K, consistent with bare rock; a **thick CO_2 atmosphere is ruled out** (Greene et al. 2023). With TRAPPIST-1 c and LHS 475 b, inner M dwarf rocky planets show a widespread lack of thick atmospheres.



JWST NIRCam and MIRI emission spectrum of 55 Cancri e: the data fall below the bare-rock blackbody and favour CO and CO_2, tentative evidence for a secondary atmosphere outgassed from a molten surface. Hu et al. (2024).



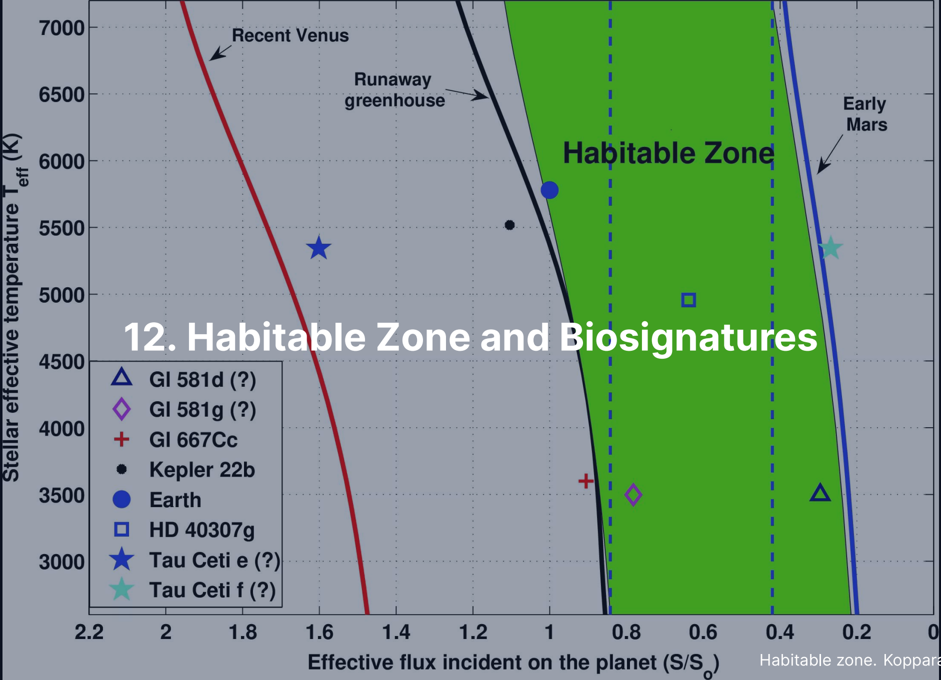
JWST NIRSpec G395H transmission spectrum of the Earth-size M dwarf planet LHS 475 b at about 12 pc: flat, so hydrogen-helium and pure CH_4 atmospheres are ruled out and a Venus-like CO_2 atmosphere stays marginally consistent. Lustig-Yaeger et al. (2023).



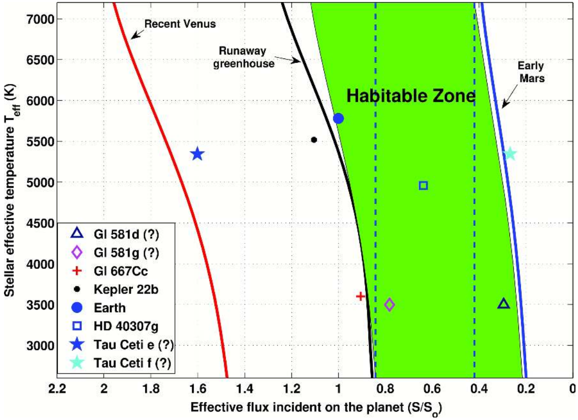
JWST transmission spectrum of the sub-Neptune K2-18 b in the habitable zone of its M dwarf. Madhusudhan et al. (2023).

$\mathrm{CH_4}$ and $\mathrm{CO_2}$ detected, DMS disputed and tentative ($\lesssim 2\sigma$); Hycean ocean, mini-Neptune envelope and magma ocean models compete: candidate biosignatures need rigorous real-time vetting.

12. Habitable Zone and Biosignatures



Habitable zone. Kopparapu et al. (2013)



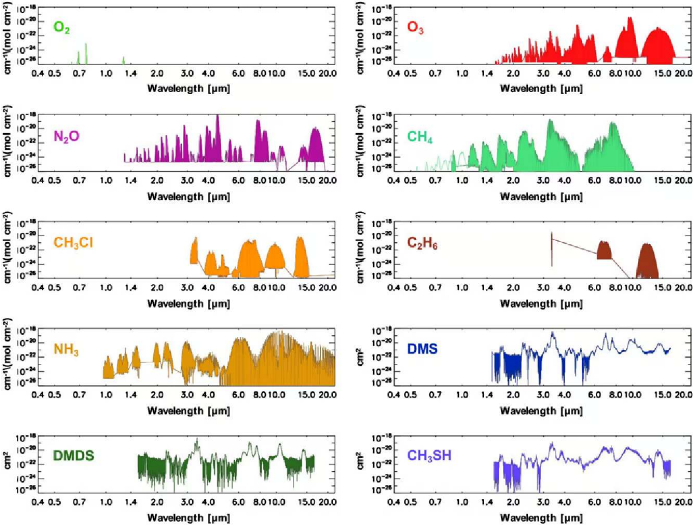
The classical habitable zone from an updated 1D radiative-convective grid. Kopparapu et al. (2013).

Inner edge: the Simpson-Nakajima runaway greenhouse; outer edge: the maximum $\mathrm{CO_2}$ greenhouse before $\mathrm{CO_2}$ condenses; both scale with stellar effective temperature and luminosity.

Habitable Zone Evolution and Biosignatures

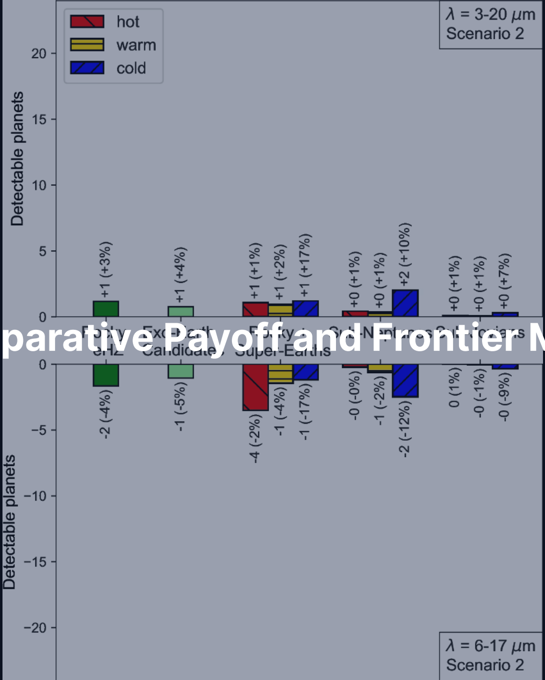
- **Dynamic trajectory:** HZ boundaries shift with stellar evolution; pre-MS M dwarfs subject planets to prolonged runaway desiccation
- **Disequilibrium pairs:** atmospheric biosignatures demand chemical disequilibrium (O_2 + CH_4), not lone gas species
- **False positives:** abiotic O_2 can build up by H_2O photolysis with hydrogen escape, or by CO_2 photolysis

Life detection is an inverse problem requiring exclusion of all abiotic pathways.

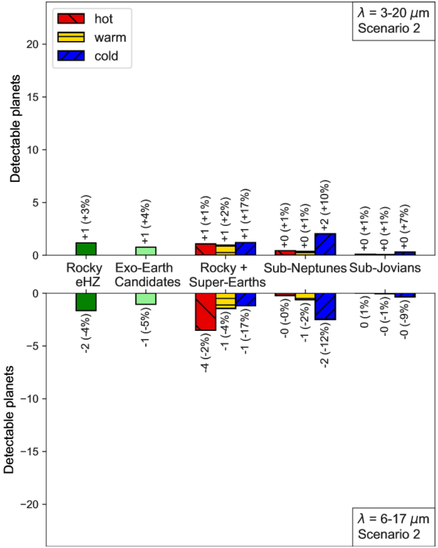


Absorption cross-sections from 0.4 to 20 μm for ten candidate biosignature gases: oxygen absorbs only in narrow optical bands, the others mostly in the infrared. Reproduced from Schwieterman et al. (2018).

13. Comparative Payoff and Frontier Missions



LIFE detection yield. Quanz et al. (2022)



Predicted detection yields of the LIFE mid-infrared space interferometer for exoplanets within 20 pc, by radius and insolation. Quanz et al. (2022).

Summary: Today's Course-Level Takeaways

1. **Detection:** transit plus radial velocity breaks $m \sin i$ and yields the true bulk density; every method carries its own bias
2. **Demographics:** small planets dominate ($\eta_\oplus \sim 0.4$); photoevaporation or core-powered mass loss carves the radius valley at $\sim 1.8\,R_\oplus$; resonant chains and peas-in-a-pod systems record smooth disk migration
3. **Characterisation:** JWST detects photochemistry and bare rocks; biosignatures require ruling out false positives

Next: Lecture 14

Questions?

Next: **Lecture 14. Synthesis & Astrobiology**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`